Teacher answers and topic guidance

Accept any scientifically equivalent wording in spoken, signed, pointed, drawn, or written form. Every question's wording is listed beside its number below, and the I do and We do model answers match the lesson page word for word.

Starter

1. A claim says: 'Clouds make night because they hide the Sun.' Why does the fixed-lamp model refute this claim?

Option A is right. The model has no clouds and the lamp stays on. Rotation turns marker X to the unlit side.

I do model answer

1. How do we record a four-frame rotation cycle? How do we explain apparent solar motion and model limits?

Follow four steps in the 24-hour rotation record. Step 1 (Fixed lamp): View Earth from above the North Pole. The fixed lamp is on the left. Track marker X through four anticlockwise quarter turns. Step 2 (Ordered frames): Frame C (Top, sunrise) enters light. The Sun appears low in the east. Frame D (Left, midday) faces the lamp. Frame B (Bottom, sunset) leaves light. Frame A (Right, night) faces away in shadow. Step 3 (Apparent motion): Earth rotates west to east. We turn with Earth, so the Sun has apparent east-to-west motion. Step 4 (Refute and limit): Night happens without clouds while the lamp stays on. Equal model frames simplify real scale, latitude, and seasonal axial tilt.

We do model answer and checkpoint

1. Inspect Frames C, D, B, and A in the 24-Hour Control Room. Match each frame to its light state. Then repair the faulty night explanation.

Frame C (Top, sunrise): marker X enters the lit half. Frame D (Left, midday): marker X faces the fixed lamp directly. Frame B (Bottom, sunset): marker X leaves the lit half. Frame A (Right, night): marker X faces away in the unlit half. Repair: do not say the lamp switched off or clouds hid the Sun. State that Earth's rotation carried marker X to the unlit side.

2. Look at the North-Pole view with the lamp on the left. Why is marker X unlit in Frame A (Right)?

Option A is right. Earth's rotation carries marker X onto the unlit side.

Arrival · Supported

1. **1. Look at Box 1 on the left of the arrival drawing. What does rotation mean: turning on an axis, or switching off a lamp?** — Rotation means turning on an axis.
2. **2. Look at Box 1 again. Marker X faces the fixed lamp on the left. Is marker X in day or night?** — Marker X is in day because it faces the fixed lamp.
3. **3. Look at Box 2 in the middle of the drawing. The lamp stayed on, and marker X turned into shadow. What moved: the Earth model or the lamp?** — The Earth model rotated while the lamp stayed fixed.
4. **4. Look at Box 3 on the right of the drawing. What completes one full rotation cycle: returning to the start, or stopping halfway?** — Returning to the starting position completes one full rotation cycle.

Arrival · Independent

1. **1. Use Box 1 of the arrival drawing. Distinguish Earth's daily rotation from Earth's yearly orbit.** — Rotation is Earth turning on its axis in 24 hours. Orbit is Earth travelling around the Sun in one year.
2. **2. Look at Box 1 and Box 2. Marker X turns from the unlit half into the lit half. Which daily moment is this?** — This moment is sunrise because marker X is entering the lit half.
3. **3. Look at Box 2. The lamp stayed on while marker X became unlit. What physical cause does this observation support?** — Earth's rotation carried marker X onto the side facing away from the lamp.
4. **4. Look at Box 3. How do we verify that we tracked the same place through one complete cycle?** — Keep one marker X through all four ordered quarter turns until it returns to Top.

Arrival · Aspire

1. **1. Use Box 1 of the arrival drawing. Why is the daily day-night cycle caused by rotation rather than Earth's yearly orbit?** — Rotation turns each place towards and away from the Sun every 24 hours. Orbit takes a full year.
2. **2. Look at Box 2 (North-Pole view, lamp on left). Marker X moves anticlockwise from Bottom to Right. Which transition occurs?** — Marker X moves from sunset (leaving light at Bottom) into night (facing away at Right).
3. **3. Why is apparent Sun motion across the sky different from the Sun orbiting Earth?** — Apparent motion describes the observer's view from a rotating Earth. The Sun does not orbit Earth.
4. **4. Look at Box 3. Do four equal model frames prove that daylight lasts 12 hours everywhere on Earth?** — No. Equal frames are a model simplification that omits latitude and seasonal axial tilt.

Arrival · Choice

1. **1. Look at Box 1 of the arrival drawing. Does rotation mean turning on an axis or staying still?** — Rotation means turning on an axis.
2. **2. Marker X faces the lit lamp on the left. Is marker X in day or night?** — It is in day because it receives light.
3. **3. Look at Box 2. Marker X turns into the dark half. Did the lamp switch off: No or Yes?** — No. The lamp stayed on while Earth rotated.
4. **4. Look at Box 3. How many quarter turns make one full cycle back to the start: four or one?** — Four quarter turns make one full cycle back to the start.

You do · Supported mark scheme

1. **1. Use Reference A. Order Frames C, D, B, and A starting at sunrise (Top). Match each frame to entering light, facing light, leaving light, or facing away.** — Order: Frame C (Top, sunrise, entering light), Frame D (Left, midday, facing light), Frame B (Bottom, sunset, leaving light), Frame A (Right, night, facing away). *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Use Reference A. Explain why Frame D is day and Frame A is night. Complete: 'Marker X has night because Earth ___ and X faces ___ from the lamp.'** — Frame D

faces the fixed lamp so it is lit. Marker X has night because Earth rotates and X faces away from the lamp. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

3. **3. Use Reference B. Refute the claim 'Clouds make night.' Then add one // improvement after adult feedback.** — The model has no clouds and the lamp stays on, yet Frame A is dark because X rotated away. // Added 'in Frame A (Right), marker X faces away from the fixed lamp'. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

You do · Independent mark scheme

1. **1. Use Reference A. Complete a four-row evidence record for marker X (Frames C, D, B, and A). State the position, illumination state, and apparent Sun position.** — Frame C (Top): entering light, Sun low in east. Frame D (Left): facing light, Sun high at midday. Frame B (Bottom): leaving light, Sun low in west. Frame A (Right): facing away, unlit night. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Use Reference A. Explain why the Sun appears to move east to west across our sky. State what moves and what stays fixed in the model.** — Earth rotates west to east (anticlockwise from above the North Pole) while the lamp stays fixed. Because we turn with Earth, the Sun has an apparent east-to-west motion. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
3. **3. Use Reference B. Use model evidence to refute 'Clouds cause night.' State one model limit, and add one // revision after feedback.** — No clouds are present and the lamp stays on, so rotation causes night. Limit: distance, size, and seasonal tilt are not to scale. // Added 'equal quarter turns do not show seasonal day length'. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

You do · Aspire mark scheme

1. **1. Use Reference A. Construct a four-frame North-Pole evidence account for marker X. Link anticlockwise rotation to apparent east-to-west solar motion.** — From above the North Pole with a fixed lamp on the left, anticlockwise (west-to-east) rotation carries marker X through C (Top, sunrise, Sun low east), D (Left, midday), B (Bottom, sunset, Sun low west), and A (Right, night). Turning with Earth makes the Sun appear to cross from east to west. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Use Reference B. Distinguish observation from inference to refute 'Clouds cause night.' Explain why a moving lamp could mimic the pattern but represent false physics.** — Observation: lamp stays fixed and on, no clouds exist, D is lit and A is unlit. Inference: rotation changes orientation to cause night. A moving lamp mimics the light pattern but falsely models a daily solar orbit. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
3. **3. Evaluate one strength and one limit of the four-frame model. Add a two-part // revision after feedback.** — Strength: tracks one location under controlled illumination. Limit: equal 90-degree frames omit 23.5-degree axial tilt and latitude differences in daylight duration. // Bounded claim to equatorial symmetry and added Frame A/D locators. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

You do · Choice mark scheme

1. **1. 1. Look at Reference A. Which frame shows midday facing the lamp: Frame D (Left) or Frame A (Right)? 2. Which frame shows night facing away: Frame A or Frame D?** — 1. Frame D (Left) shows midday facing the lamp. 2. Frame A (Right) shows night facing away. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. 3. What moves in this model: rotating Earth or the fixed lamp? 4. Why does the Sun seem to cross our sky: we rotate or Sun stops?** — 3. Rotating Earth moves while the lamp stays fixed. 4. We rotate with Earth. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*
3. **3. 5. Look at Reference B. Does the model need clouds to make night: No or Yes? 6. Name one model limit: not to scale or real space size? Add one // edit.** — 5. No, rotation makes night without clouds. 6. The model is not to scale. // Added 'Frame A is unlit because X faces away'. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

Exit ticket · Supported

1. **1. Look at the North-Pole model. Which frame is in night: Frame D (Left) or Frame A (Right)?** — Frame A (Right) is in night because marker X faces away from the lamp.
2. **2. Does the model need clouds to make night? Give one clue from the model.** — No. There are no clouds and the lamp stays on while Earth rotates.
3. **3. How do we mark an improvement in our book after feedback?** — Keep the first draft visible and write // before the improvement.

Exit ticket · Independent

1. **1. Use Frame D (Left) and Frame A (Right) to explain how day changes into night. State what moved and what stayed fixed.** — Earth rotated marker X from the lit left side (Frame D) to the unlit right side (Frame A) while the lamp stayed fixed and on.
2. **2. Why does the Sun appear to move east to west each day?** — Earth rotates from west to east, so an observer turning with Earth sees the Sun appear to move from east to west.
3. **3. State one scientific limit of the four-frame classroom globe and lamp model.** — Size and distance are not to scale, and equal frames do not show seasonal day-length changes from axial tilt.

Exit ticket · Aspire

1. **1. Use Frames D and A to refute 'Clouds cause night.' Separate observation from inference.** — Observation: the lamp stays on with no clouds while D is lit and A is unlit. Inference: planetary rotation causes night by turning X away from sunlight.
2. **2. A moving torch around a still globe gives a similar light pattern. Why is that physical cause wrong?** — Both models change which side is lit, but a moving torch models a daily solar orbit instead of Earth's axial rotation.
3. **3. What real daylight question cannot be answered by four equal 90-degree North-Pole frames alone?** — They cannot show how daylight hours vary by season and latitude due to Earth's 23.5-degree axial tilt.

Exit ticket · Choice

1. **1. Why is Frame A (Right) dark: X rotated away, or the lamp switched off?** — Marker X rotated away from the fixed lamp.
2. **2. Does the lamp move or stay fixed in our model: stay fixed or orbit?** — The lamp stays fixed and on.
3. **3. Is a classroom globe and torch to real scale: No (a limit) or Yes?** — No. That is a limit of the classroom model.

Exit pupil voice

1. What helped you today, and what would you change? — Accept any pointed, spoken, signed, drawn, written, or chosen reflection on what helped and what could change next time. A pupil may pass.

Topic guidance and Key Stage 3 progression

- **Four-frame rotation evidence at Key Stage 3:** Viewed from above the North Pole with a fixed lamp on the left, Earth rotates anticlockwise (west to east), carrying marker X through **Frame C (Top, sunrise)**, **Frame D (Left, midday)**, **Frame B (Bottom, sunset)**, and **Frame A (Right, night)**.
- **Apparent motion and refuting misconceptions:** Turning west to east with Earth makes the Sun appear to move east to west. Because the lamp stays on and no clouds are present, Frame A proves night is caused by planetary rotation away from sunlight.
- **Model limitations:** Classroom globes and lamps are not to real scale, and four equal quarter-turn frames simplify latitude and Earth's 23.5° seasonal axial tilt.

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